

Project Report

Community Road Pothole Reporting System

1. Project Abstract

The **Community Road Pothole Reporting System** is a localized digital solution designed to streamline the identification and repair of road damages. By leveraging crowdsourced data, the application allows citizens to act as "sensors" for the city, providing real-time geographical data and visual evidence of potholes to help municipal authorities prioritize maintenance work effectively.

2. Key Features

- **Interactive Reporting:** Multi-input form for location, severity, and detailed descriptions.
- **Geospatial Mapping:** Real-time rendering of reported issues on an interactive map using Leaflet.js.
- **Visual Documentation:** Support for image uploads to provide verifiable evidence of road conditions.
- **Prioritization Logic:** An upvoting system to gauge community impact and urgency.
- **Analytical Dashboard:** Automated status tracking and visualization via doughnut charts.

3. Technical Specifications

Category	Technology Stack / Tool
Frontend Structure	HTML5, CSS3 (Custom Responsive Layout)
Dynamic Logic	Vanilla JavaScript (ES6+)
Mapping API	Leaflet.js & OpenStreetMap
Data Visualization	Chart.js
Data Persistence	Browser LocalStorage API

4. System Architecture & Workflow

A. User Submission Phase

Users interact with a clean, responsive interface to submit pothole details. The application uses the `FileReader` API to process image uploads locally without needing a backend server for the initial prototype.

B. Data Synchronization

All reports are serialized into JSON and stored in `localStorage`. This ensures that data persists across browser sessions, allowing the community to build a cumulative database of road issues.

C. Real-time Visualization

The map logic generates dynamic markers based on coordinates (simulated for the prototype). Simultaneously, `Chart.js` aggregates data to show the ratio of "Open" vs "Resolved" cases, providing a high-level overview of progress.

5. Conclusion

This project demonstrates the power of community-driven data. By digitizing the reporting process, the Community Road Pothole Reporting System reduces the response time for infrastructure repairs and fosters a sense of civic responsibility among residents.